

Final Project

Phase II - Decision Making

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IFT 533: Data Visualization & Reporting for IT

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Section 1: Used Visualization Tools

Tableau was selected as the visualization tool for the dashboard due to several compelling reasons:

Data Size and Complexity Handling: Given that our dataset is of medium size, Tableau is well-suited to manage it effortlessly. Unlike tools designed for high-dimensional datasets, Tableau provides an optimal solution for our specific dataset.

Customization Capabilities: Tableau offers a high degree of customization, enabling the creation of visualizations tailored to meet specific needs and objectives. This adaptability is particularly beneficial for small to medium-sized businesses with unique data visualization requirements that may not be fulfilled by off-the-shelf solutions.

User-Friendly Interface: Tableau is recognized for its user-friendly design, setting it apart from other visualization tools like Python Bokeh. Its drag-and-drop interface facilitates the creation of interactive visualizations without the need for extensive coding, making it accessible to users with varying levels of technical expertise.

Ease of Use: The intuitive nature of Tableau allows users to focus on formulating questions and refining visualization ideas, eliminating the need for intricate coding. Additionally, Tableau's provision of pre-built visualizations enhances flexibility, enabling the creation of compelling visuals with minimal prior experience.

Section 2: Explanation of Required Data Pre-processing

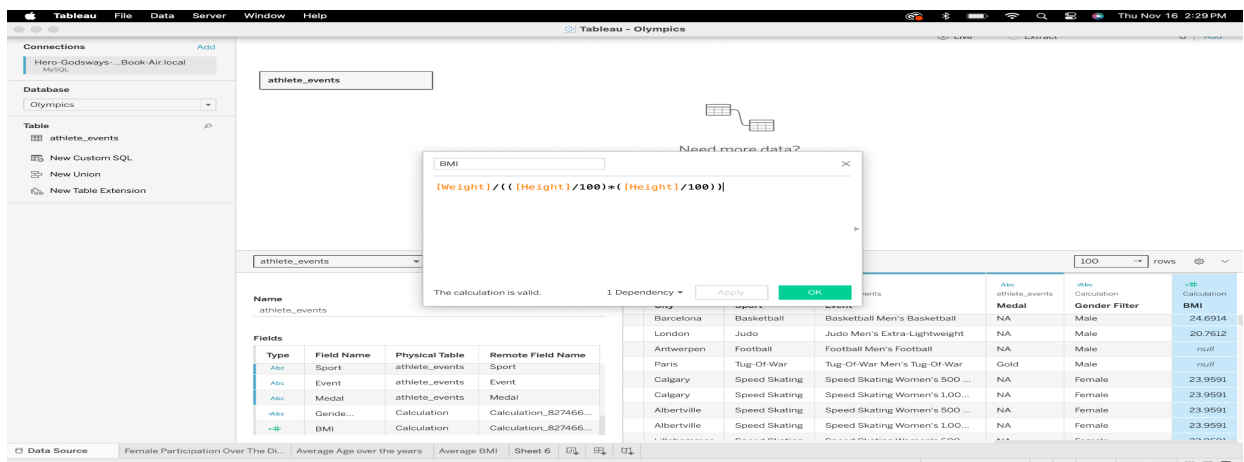
The selected dataset underwent necessary pre-processing to make it suitable for visualization, and Tableau and Python were employed as the tool for this data preparation.

Creating BMI Column using Calculated Field:

To compute the BMI of participating athletes, data preprocessing was computed using Tableau. The data from individual athlete parameters such as weight and height were combined using the formula $\text{Weight}/(\text{height})^2$.

Preprocessing steps

1. Convert the height from centimeters to meters by dividing by 100. ($\text{height}/100$).
2. Divide the recorded athlete weight by the square of the height (kg/m^2)
3. Preprocessing Calculated Field



Creating a column of Male Dominated Sports using a Calculated Field

To compute the sports that have been male dominated in the history of the olympics, data preprocessing was done to streamline male prevalent sports from other sports, in order to determine the increase in female participation over the years.

Male Dominated Field

×

```
IF [Sport] = 'Gymnastics' OR
   [Sport] = 'Bobsleigh' OR
   [Sport] = 'Canoeing' OR
   [Sport] = 'Volleyball' OR
   [Sport] = 'Judo' OR
   [Sport] = 'Athletics' OR
   [Sport] = 'Archery' OR
   [Sport] = 'Diving' OR
   [Sport] = 'Basketball' OR
   [Sport] = 'Hockey' OR
   [Sport] = 'Shooting' OR
   [Sport] = 'Alpine Skiing' OR
   [Sport] = 'Fencing' OR
   [Sport] = 'Rowing' OR
   [Sport] = 'Sailing' OR
   [Sport] = 'Wrestling' OR
   [Sport] = 'Boxing'
THEN 'Male-Dominated'
ELSE 'Other Sports'
END
```

The calculation is valid.

1 Dependency ▾

Apply

OK

The calculated field makes all the olympic sports into two categories “male-dominated” and “other sports”, thereby allowing us to differentiate and analyze the trend in either growing or declining participation for either gender.

Section 3: Lists of Final set of Potential Questions

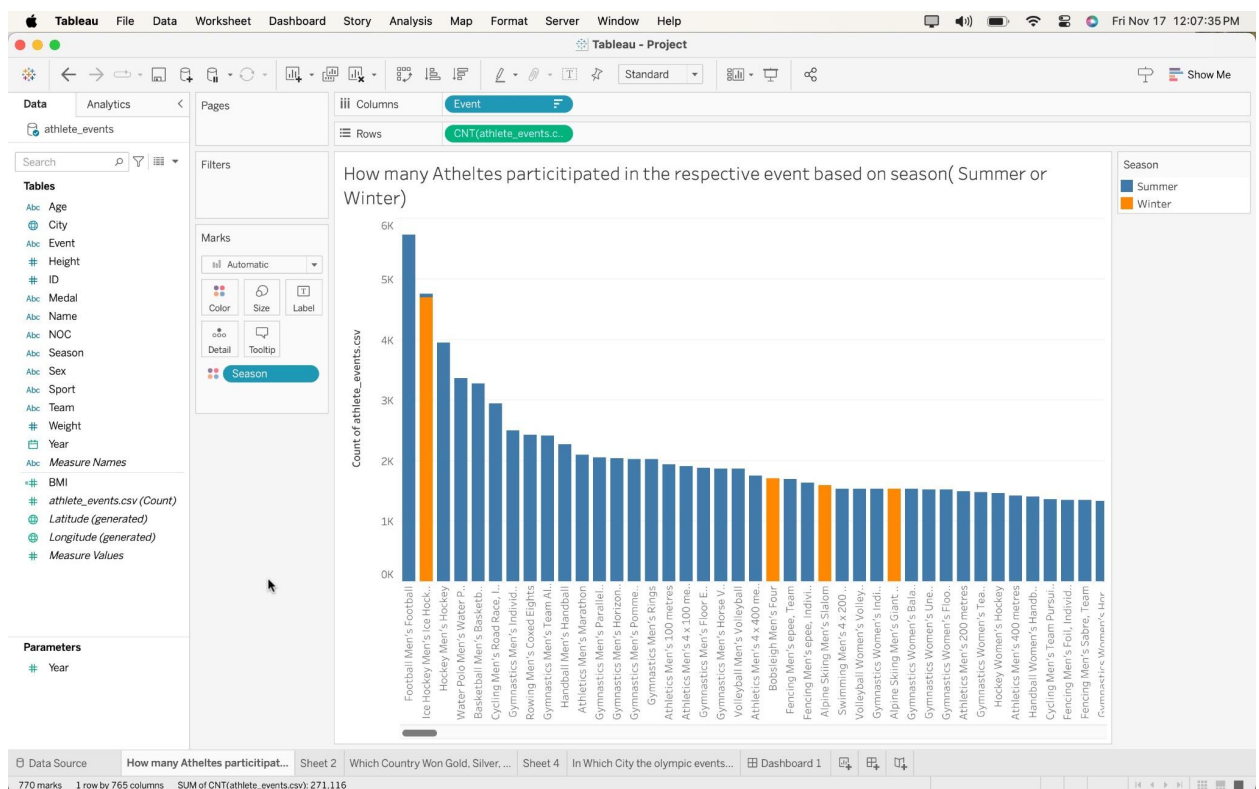
1. How many Athletes participated in the respective event based on season(Summer or Winter)
2. In the past 120 years of Olympic events, what is the count of gold, silver, bronze for each country in different years and overall years?
3. For each host country, what is the count of different medals achieved by the different teams.
4. Has there been variations in female participation over the years particularly in often male dominated sports.
5. What is the distribution of the age of gold medalists in either the summer or winter seasons?
6. What is the global distribution of medals across regions?
7. In the individual sports categories, where has the United States struggled to achieve medals?
8. What is the variations for female participation for the different seasons over the years
9. For selected teams(countries), what is the average age of women and men participating in the Olympics over the years
 - a. How were the change in average age affected by global events such as world war I & II or the pandemics?
10. For the different sports, how has the average BMI of winners changed over the past years compared to non-winning participants?

Section 4: Dashboard Plots Draft

1. How many Athletes participated in the respective event based on season(Summer or Winter)

This visualization can represent how many athletes have participated in different seasons. In such a way, we plotted a stacked bar graph which represents the seasons of each bar with two different colors. To make this visualization possible, we considered **Event** attribute and **CNT(Athlete_Events)** which is the count of total athletes in the dataset. Finally Seasons for getting different colors while stacked. The Pre-Attentive attributes that we can identify are the length and color.

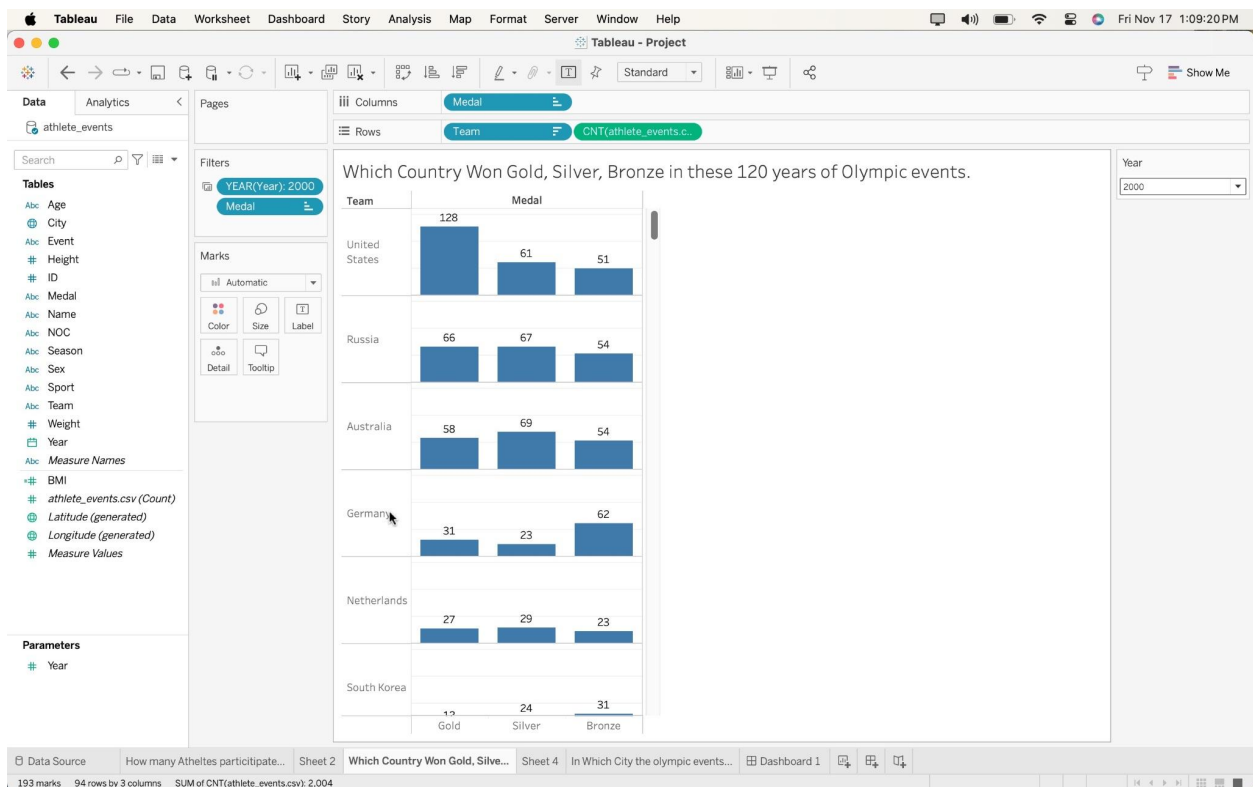
- **Length:** This defines the total number of athletes participating in each event.
- **Color:** Different seasons in bars are represented with different colors. Here, Blue refers to Summer and Orange represents Winter.

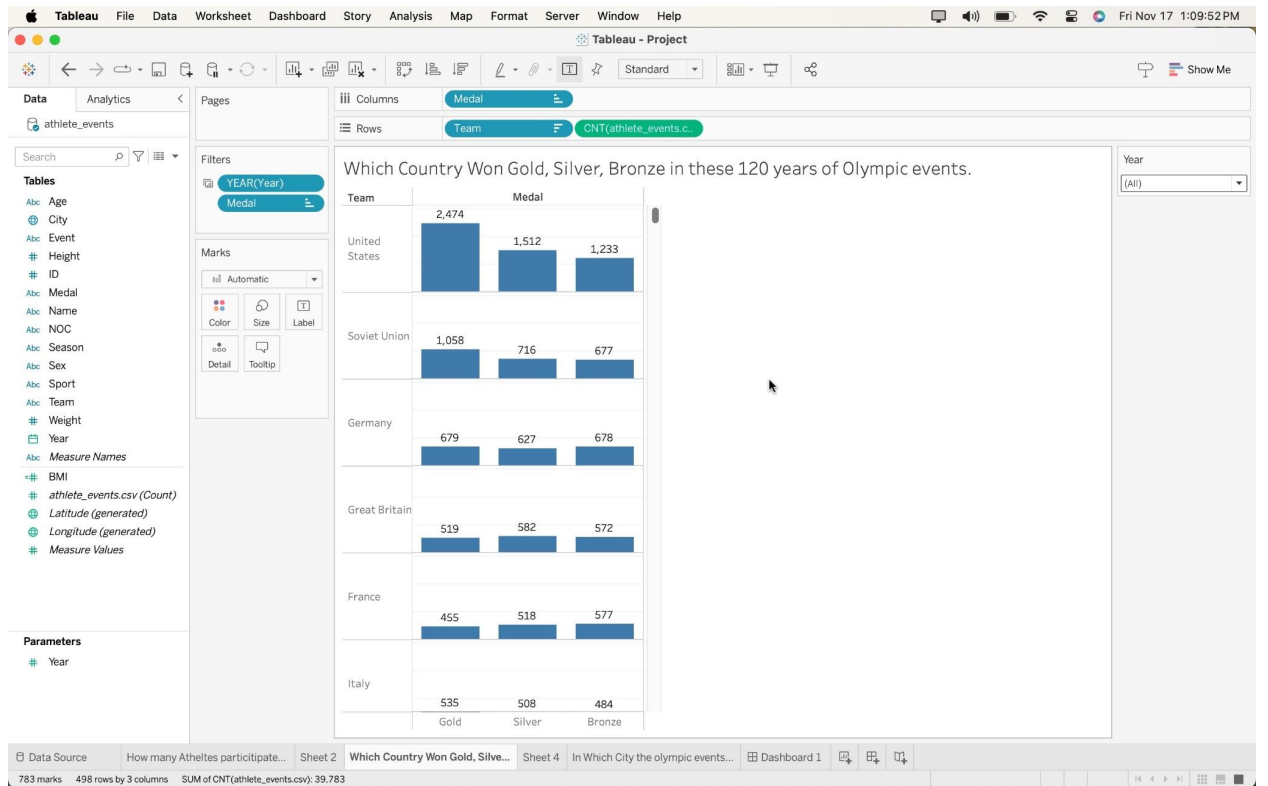


2. In the past 120 years of Olympic events, what is the count of gold, silver, bronze for each country in different years and overall years?

This visualization shows the country medals earned in different years. By this way we can visualize how each country got more medals in the different years. To make this visualization possible, I used a multi series bar chart where it can show each country with their medals won in different years or overall years. To achieve this I considered **Medals** in columns section and excluded the **Not Achieved** any medals by Athletes. In row, I selected **Team** and **CNT(Athlete_Events)**. Also used **Year** as filter to get the medals in respective years by the Teams. The Pre-Attentive attribute that we can identify is length .

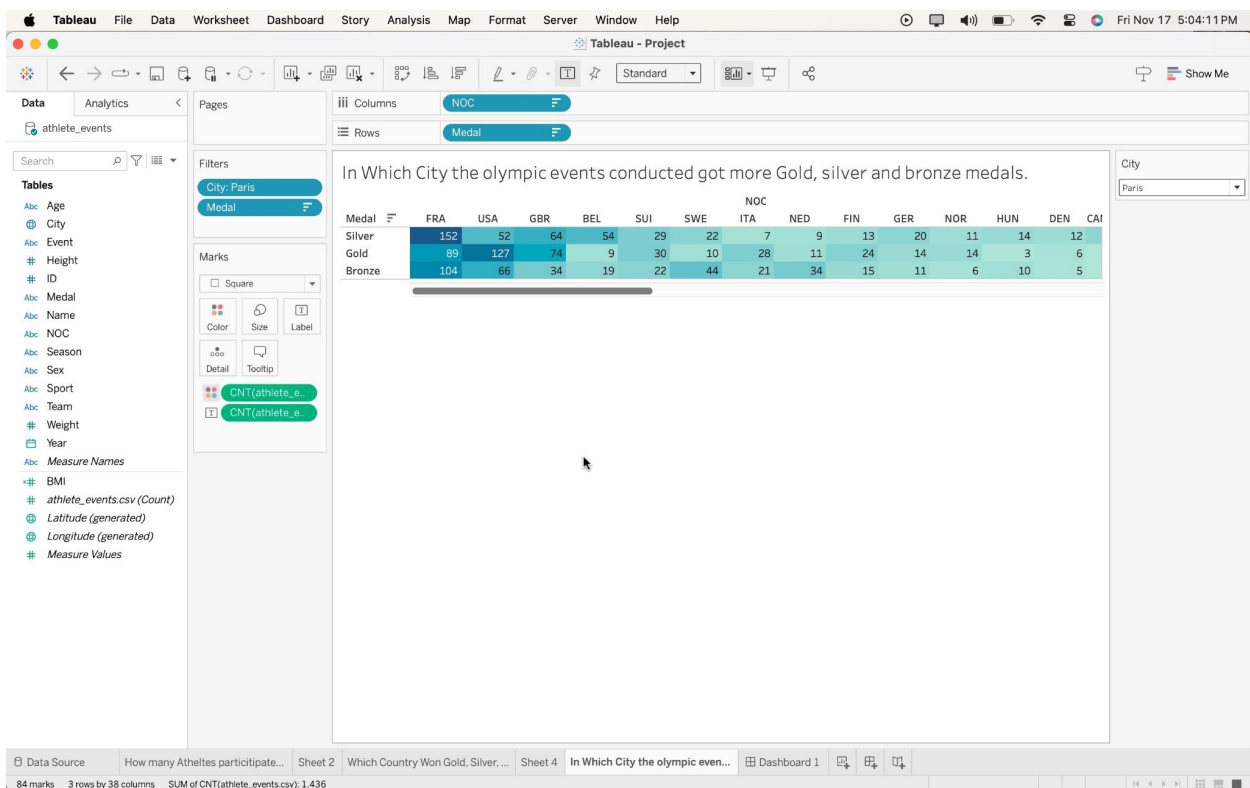
- **Length:** This defines the total number of medals achieved by each team in different years.





3. For each host country, what is the count of different medals achieved by the different teams.

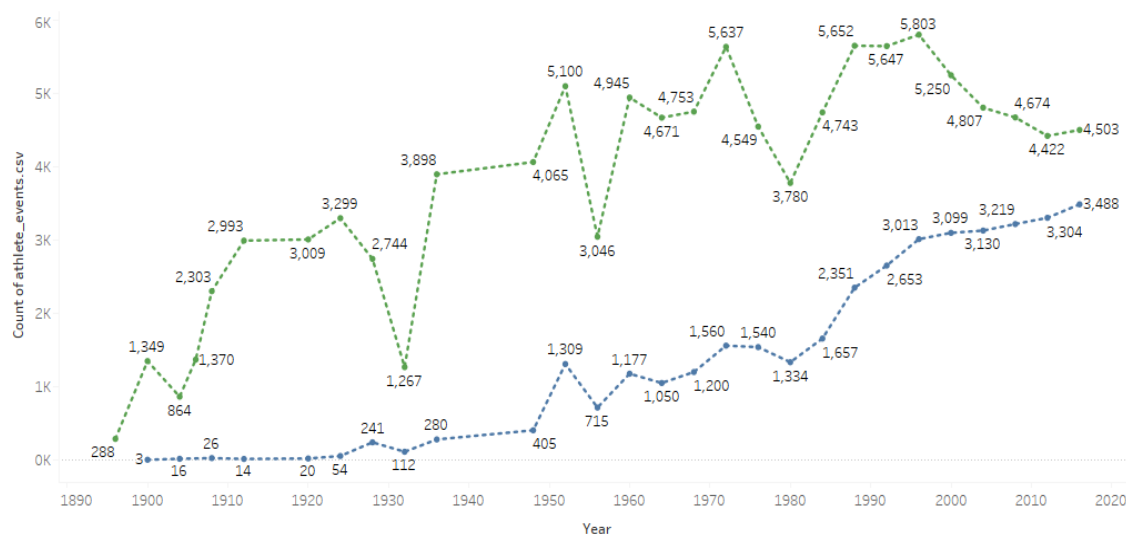
In this visualization, we are trying to show the medals achieved by the different teams based on Olympic events hosted by countries in their major cities. This visualization is shown in Table format having 3 rows, Gold, Silver and Bronze (Excluded Not Achieved Athletes). And the Columns of the country got medals. Using the **City** as a filter, we can see how many medals were won while the Olympic event is hosted in the city.



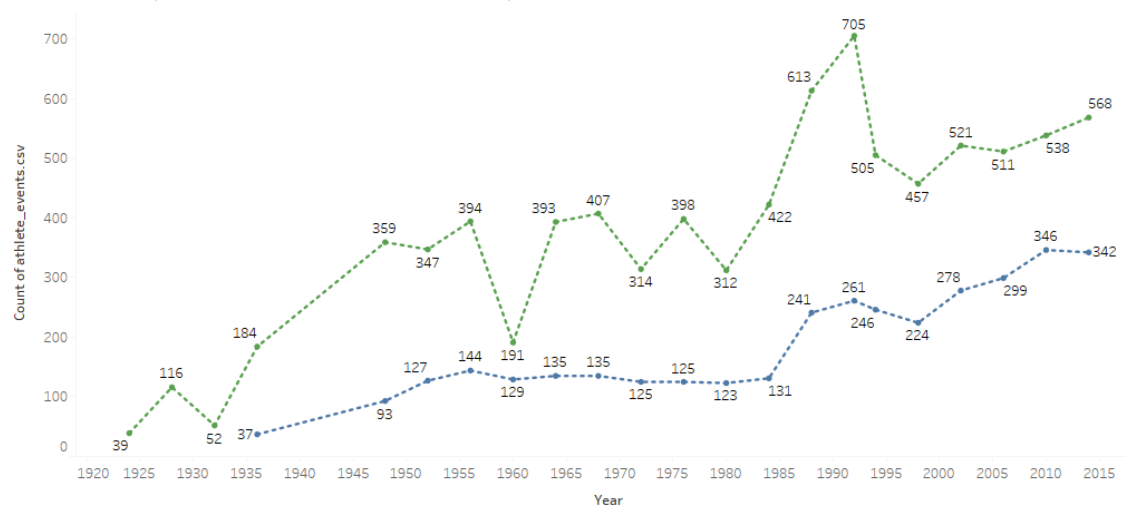
4. Has there been an increase in female participation over the years particularly in often male dominated sports?

This visualization tracks the rise of female participation in the Olympics, specifically in traditionally male-dominated sports. Using a two-column table, it showcases yearly data, distinguishing between male and female participants. A color scheme or shape helps differentiate genders, while a filter isolates male-dominated sports for focused insights.

Female Participation count in Male Dominated Sports



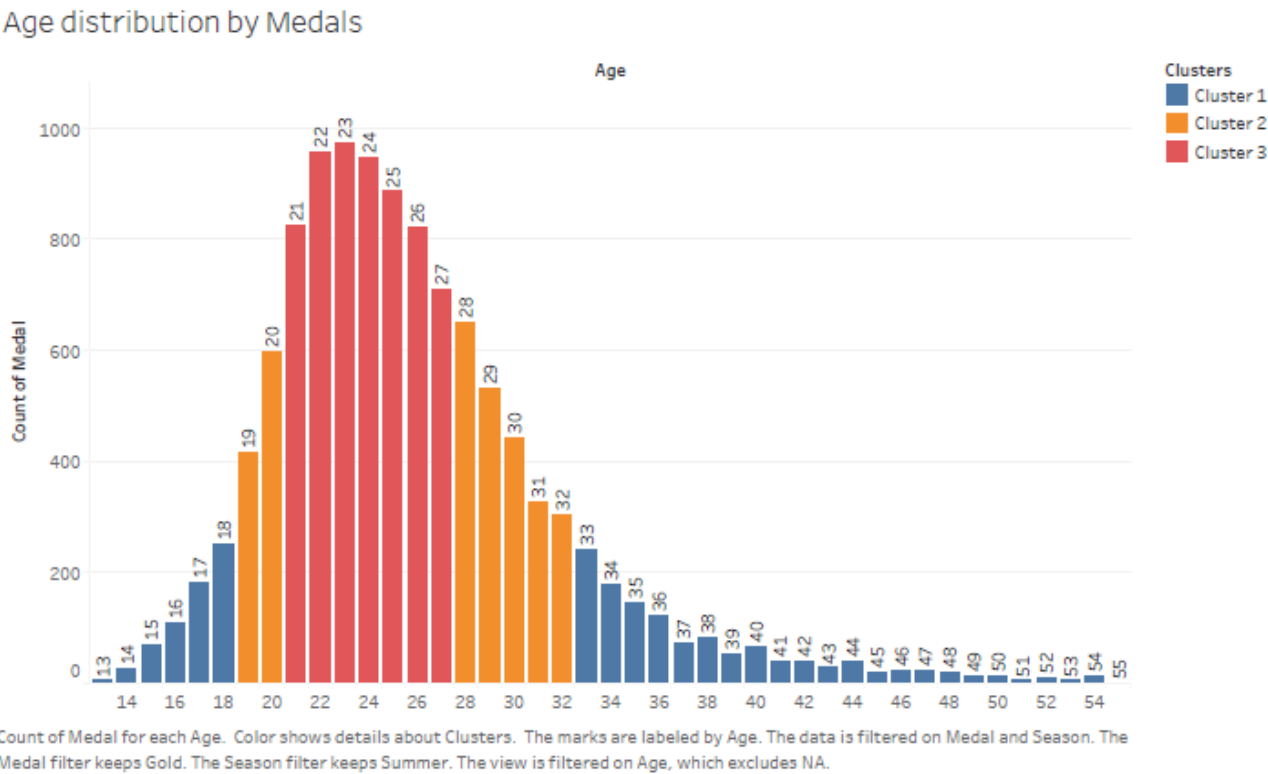
Female Participation count in Male Dominated Sports



The goal is to provide a comprehensive view of the evolving landscape of female participation in the Olympics. The trend of count of athlete_events.csv for Year. Color shows details about Sex. The marks are labeled by count of athlete_events.csv. The data is filtered on Male Dominated Field and Season. The Male Dominated Field filter keeps Male-Dominated. The Season filter keeps Winter. The view is filtered on Sex, which keeps multiple members.

5. What is the distribution of the age of gold medalists in either the summer or winter seasons?

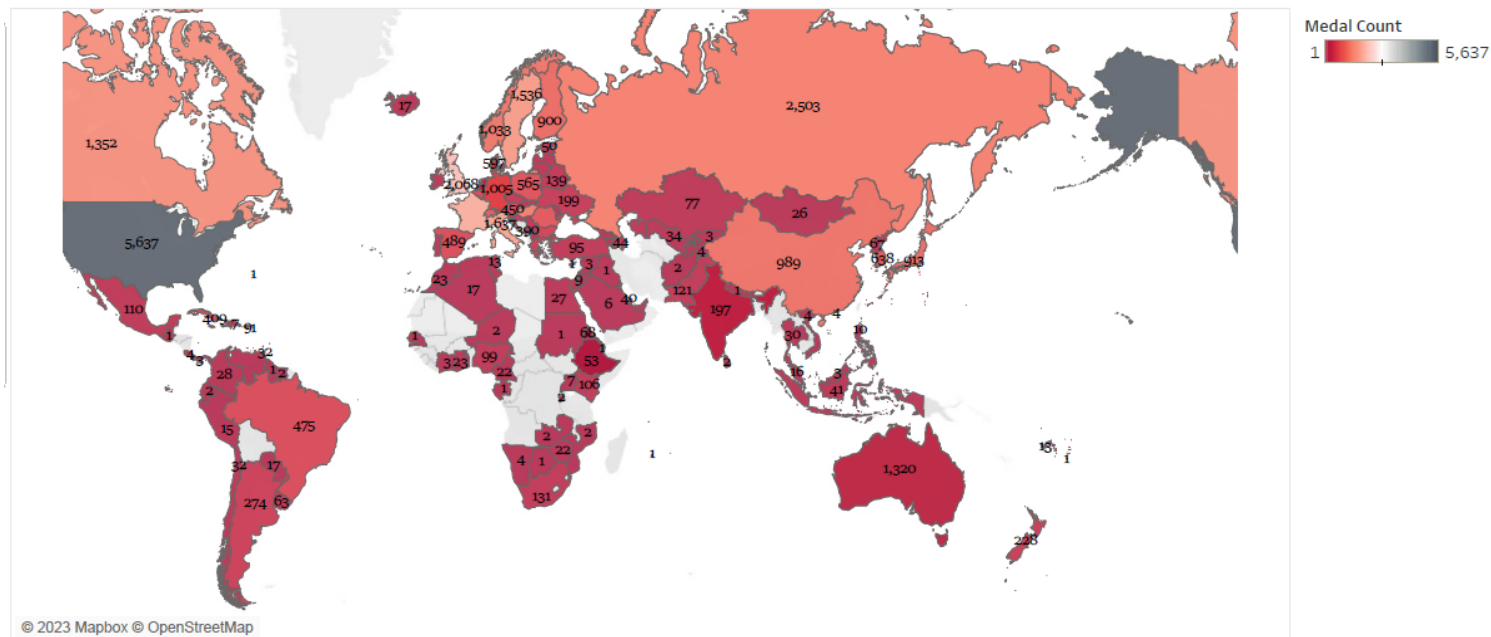
This visualization illustrates the age distribution of gold medalists, differentiating between summer and winter seasons. Using a two-column structure, it displays age groups with distinct visual cues. A filter allows users to focus on either season



6. How evenly are the acquired medals distributed on a global scale?

This visualization showcases the global distribution of Olympic medals won over the 120-year history of the Games and analyzes whether there has been a fair spread or distribution.

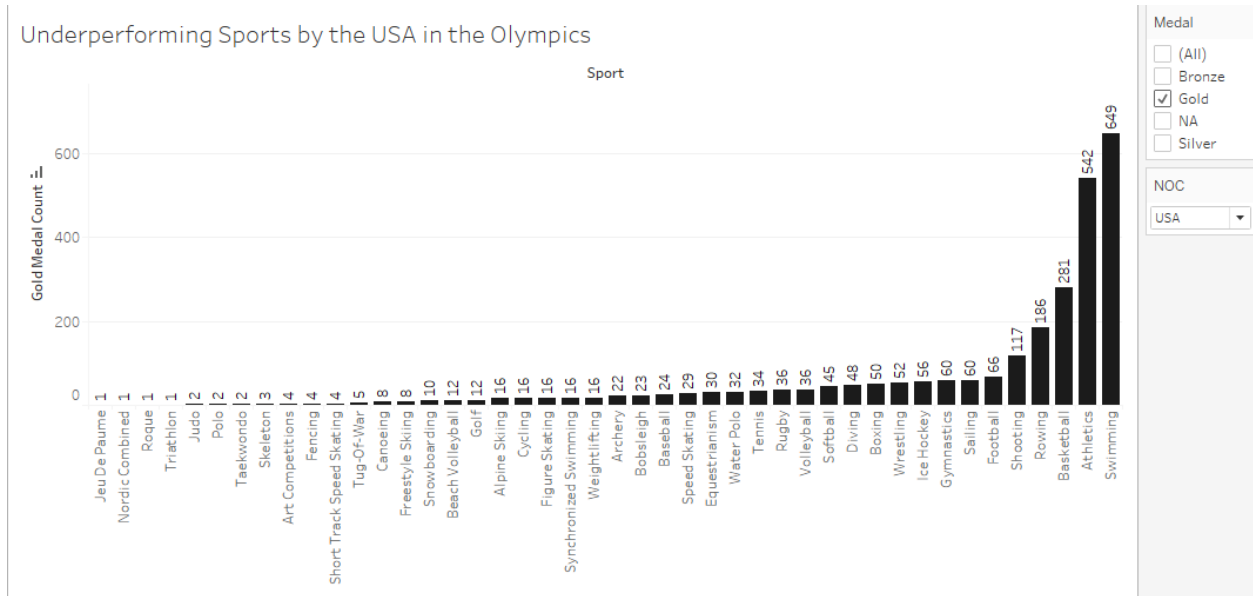
Global Medal Distribution



Map based on Longitude (generated) and Latitude (generated). Color shows sum of Medal Count. The marks are labeled by sum of Medal Count. Details are shown for NOC. The data is filtered on Medal, which keeps Bronze, Gold and Silver.

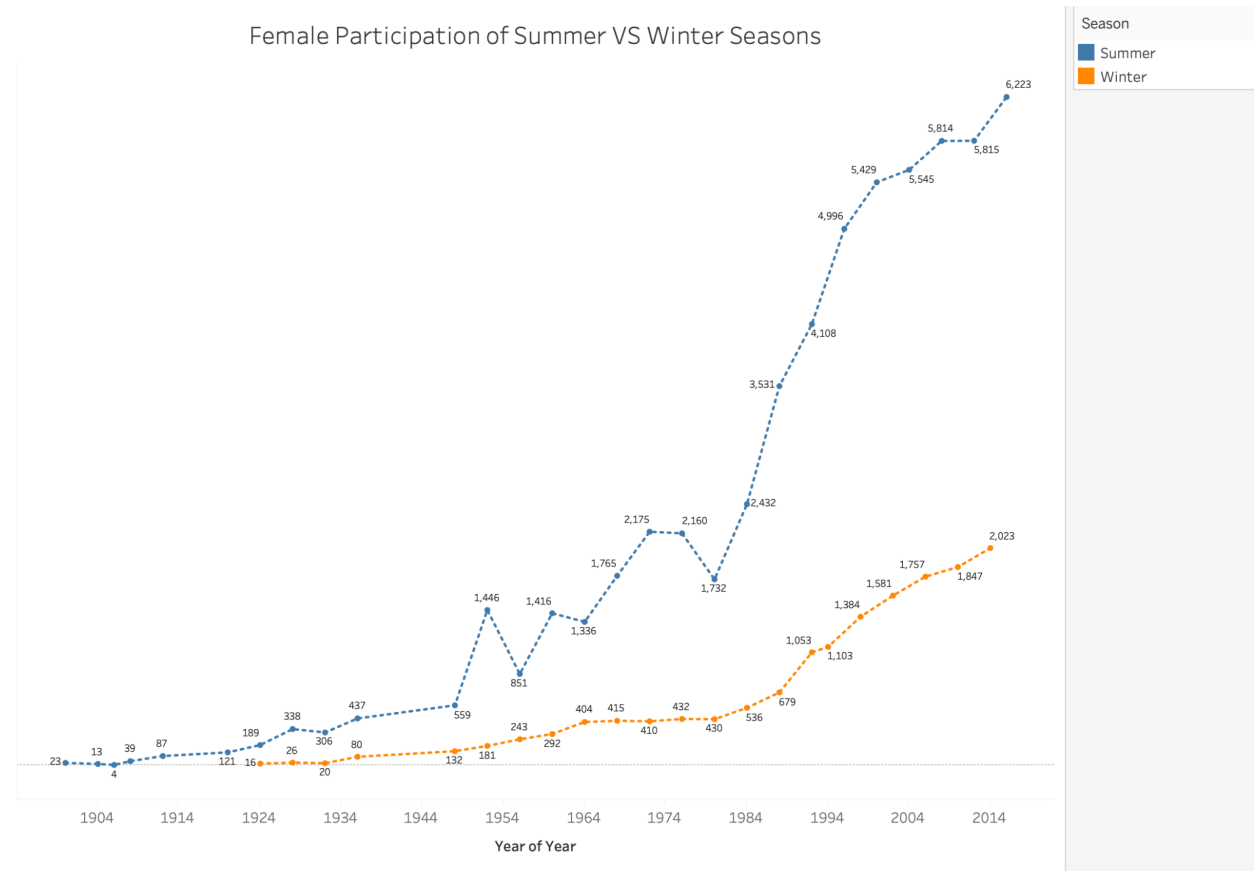
7. In which Olympic sport or discipline has the United States achieved the fewest gold medals over the course of the 120-year history of the Games?

Sum of Gold Medal Count for each Sport. The marks are labeled by sum of Gold Medal Count. The data is filtered on Medal and NOC. The Medal filter keeps Gold. The NOC filter keeps the USA.



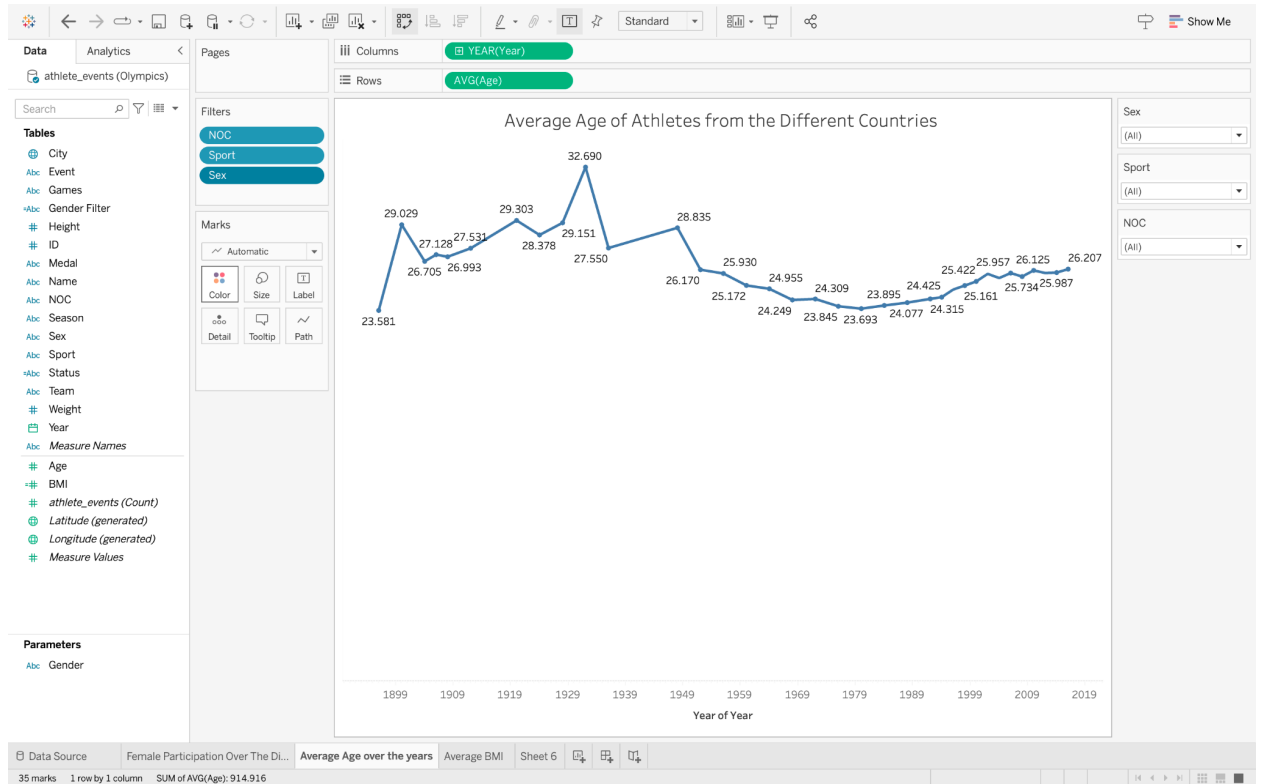
8. What is the variations for female participation for the different seasons over the years

The chart shows a growing female participation rates in the olympic season with more women involved in the summer sports compared to the winter sports.



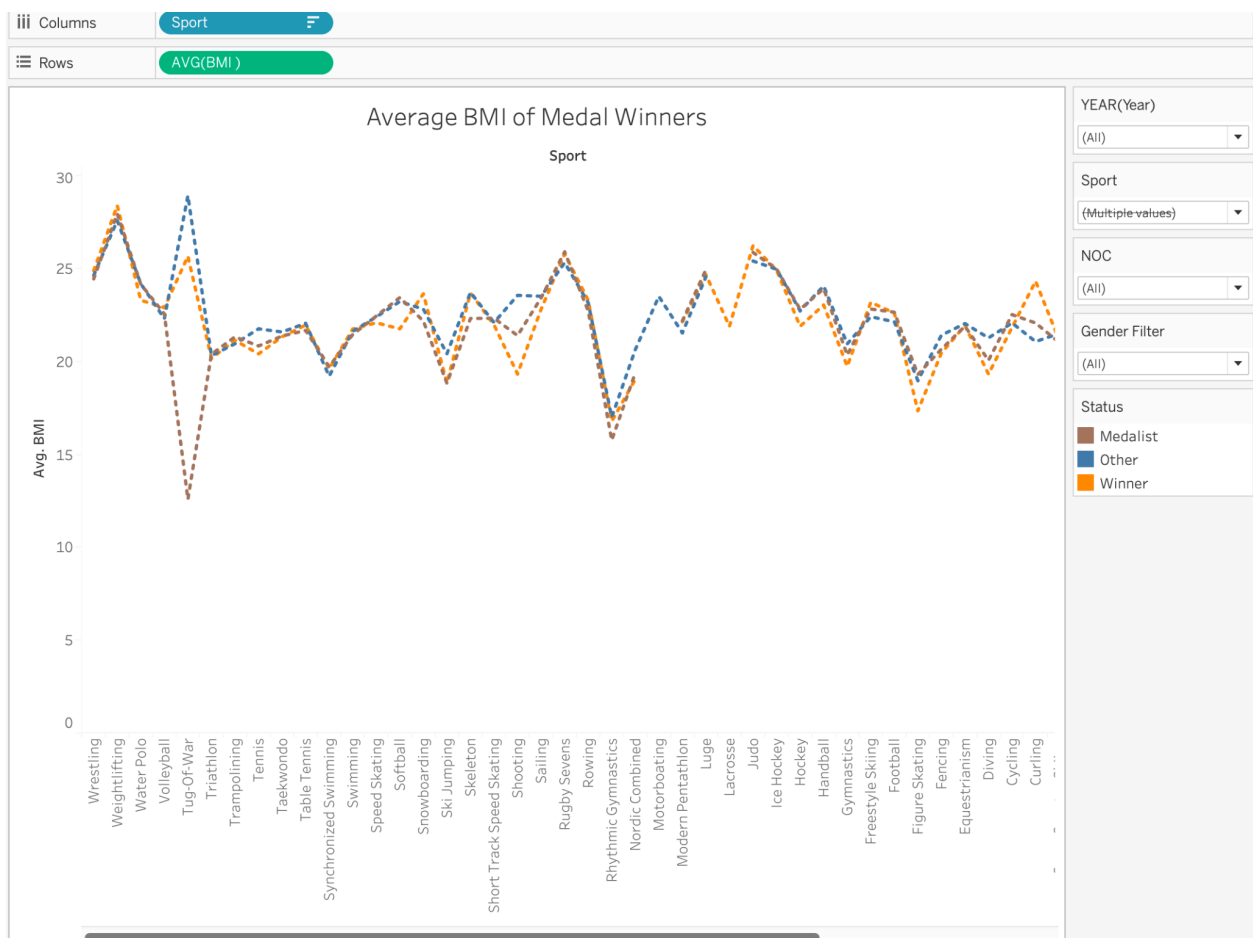
9. For selected teams(countries), what is the average age of women and men participating in the Olympics over the years

- a. The line chart below was chosen to provide a clean overview of how the average age of participants has changed over the years. The audience are able to compare different sexes, countries, and also compare how historic events such as the world wars may have affected the average age of participants.



10. For the different sports, how has the average BMI of winners changed over the past years compared to non-winning participants?

This chart shows how the average BMI of participants have changed over the years for the different sports. Allowing the audience to filter by sports, country, and year. The audience are also able to compare the BMI of the respective winners, other medalists and all other participants. We excluded the sport who are not having the height and weight values. We have not removed the data because few athletes have won the medals. For that reason we haven't removed it because all other questions may have effects

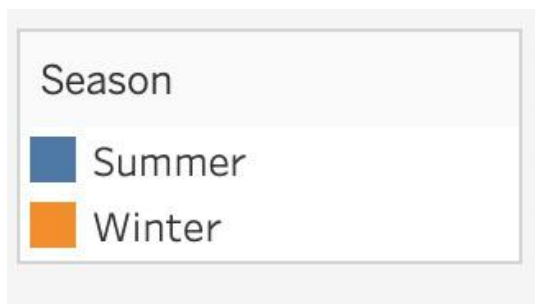


Section 5: Dashboard Interactivity

Tableau users enjoy a range of customization options to tailor their dashboards. Parameters, for instance, play a pivotal role in seamlessly transitioning between metrics, managing sorting preferences, and executing complex computations. By adjusting parameters like numerical values or computed fields, users can meet their specific requirements, influencing various aspects of worksheet operations, filters, and reference lines. The utilization of parameters, calculated fields, and filters allows for the integration of interactive controls in visualizations, providing a versatile approach to data analysis and presentation.

Seasons:

The "Season" attribute distinguishes between two distinct seasons in the visualization, aiding the identification of events based on the respective season for addressing question 1. The stacked bar chart within the visualization effectively portrays both the summer and winter seasons, visually represented by two distinct colors. This visual representation provides clarity and allows users to discern the occurrence of events in each season. The utilization of a stacked bar chart enhances the visual appeal and facilitates a comprehensive understanding of the distribution of events across summer and winter seasons.



Year:

The year filter serves as a crucial tool for pinpointing the teams or countries that secured medals in specific Olympic years. This filter is conveniently presented in a single dropdown list, allowing users to navigate through various years effortlessly. By leveraging this filter, users can gain insights into the achievements of different countries across different Olympic events in respective years. It provides a comprehensive view of how teams demonstrated their prowess in various events, showcasing their talents and accomplishments over the years. The user-friendly dropdown feature enhances the accessibility of

information related to medal-winning performances, fostering a deeper understanding of each country's Olympic journey throughout the years.

Year

(All)

▼

Enter search text

(All)

1896

1900

1904

1906

1908

1912

1920

1924

1928

1932

1936

1948

1952

1956

1960

1964

1968

1972

1976

1980

1984

1988

City:

The city filter serves as a valuable tool for pinpointing the teams or countries that secured medals in Olympic events hosted in a specific city. This filter is conveniently presented in a single dropdown list, allowing users to select the desired city of interest. By utilizing this filter, users can gain insights into the achievements and medal standings of various countries in the context of the chosen host city. The interactive nature of the dropdown list enhances the user experience, enabling them to explore and analyze Olympic data based on specific geographic locations. This feature proves particularly useful for understanding the performance and success of nations in different Olympic venues.



The image shows a web interface for a city filter. At the top, there is a label "City" followed by a dropdown arrow icon. Below this is a search input field containing the text "Paris". Underneath the input field is a search bar with the placeholder text "Enter search text". A list of cities is displayed below the search bar, starting with "(All)" and followed by various Olympic host cities. The city "Paris" is highlighted with a grey background.

City
(All)
Albertville
Amsterdam
Antwerpen
Athina
Atlanta
Barcelona
Beijing
Berlin
Calgary
Chamonix
Cortina d'Ampezzo
Garmisch-Partenkirchen
Grenoble
Helsinki
Innsbruck
Lake Placid
Lillehammer
London
Los Angeles
Melbourne
Mexico City
Montreal
Moskva
Munich
Nagano
Oslo
Paris
Rio de Janeiro
Roma

Country :

Interactive filters such as country are helpful to enable users to track the performance of a particular country of interest or compare trends with other countries. For instance users could compare two developing countries or economically different countries to analyze how the economic difference may have been reflected in sports.



The image shows a user interface for a 'Country' filter. At the top, there is a label 'Country' with a funnel icon and a dropdown menu currently set to '(All)'. Below this is a search bar with the placeholder text 'Enter search text'. A scrollable list of countries follows, each preceded by a checked checkbox. The countries listed are: (All), AFG, AHO, ALB, ALG, AND, ANG, ANT, ANZ, ARG, ARM, ARU, ASA, AUS, AUT, AZE, BAH, BAN, BAR, BDI, BEL, BEN, BER, BHU, BIH, BIZ, and BLR.

Country	Selected
(All)	☒
AFG	☒
AHO	☒
ALB	☒
ALG	☒
AND	☒
ANG	☒
ANT	☒
ANZ	☒
ARG	☒
ARM	☒
ARU	☒
ASA	☒
AUS	☒
AUT	☒
AZE	☒
BAH	☒
BAN	☒
BAR	☒
BDI	☒
BEL	☒
BEN	☒
BER	☒
BHU	☒
BIH	☒
BIZ	☒
BLR	☒

Sports:

For most fans, the choice of sports they follow are influenced by personal interest. Some of such differences are also recorded across regions with some sports fans following soccer while other may prefer to view the data related to basketball. Accordingly, visualizations depicting average BMI and average age of participants and winners also allow audiences to filter by selecting their preferred sports. This interactive filter is designed for use not only by the fans, but also prospective olympic participants, as such, the filter allows the user to select more than one sport, and to see how these values differ across. This will enable a prospective athlete hoping to participate in more than one event to consider these values carefully during the training preparation phase.

The image shows a 'Sport' filter dropdown menu. At the top, there is a search bar labeled 'Enter search text'. Below it, a list of sports is displayed, each preceded by an unchecked checkbox. The sports listed are: (All), Aeronautics, Alpine Skiing, Alpinism, Archery, Art Competitions, Athletics, Badminton, Baseball, Basketball, Basque Pelota, Beach Volleyball, Biathlon, Bobsleigh, Boxing, Canoeing, Cricket, Croquet, Cross Country Skiing, Curling, Cycling, Diving, Equestrianism, Fencing, Figure Skating, Football, and Freestyle Skiing.

Gender:

The gender filter primarily provides opportunity for users of the dashboard who maybe interested in comparing the different values to understand how the visualizations change with different sexes of the participants in the olympic games.

The image shows a 'Gender Filter' dropdown menu. It features a search bar at the top. Below the search bar, a dropdown menu is open, displaying three options: (All), Female, and Male. The (All) option is currently selected and highlighted with a grey background.

Male Dominated Field Parameter:

This sports filter adds another layer of specificity and interactivity to our dashboard. It introduces a layer of differentiation between sports that have been predominantly male centric over those that have not in the 120 year history of the olympics. It allows us to filter the visualizations for male dominated sports vs other sports and analyze trends of either genders based on their seasons of competition.

Male Dominated Field

×

```
IF [Sport] = 'Gymnastics' OR
[Sport] = 'Bobsleigh' OR
[Sport] = 'Canoeing' OR
[Sport] = 'Volleyball' OR
[Sport] = 'Judo' OR
[Sport] = 'Athletics' OR
[Sport] = 'Archery' OR
[Sport] = 'Diving' OR
[Sport] = 'Basketball' OR
[Sport] = 'Hockey' OR
[Sport] = 'Shooting' OR
[Sport] = 'Alpine Skiing' OR
[Sport] = 'Fencing' OR
[Sport] = 'Rowing' OR
[Sport] = 'Sailing' OR
[Sport] = 'Wrestling' OR
[Sport] = 'Boxing'
THEN 'Male-Dominated'
ELSE 'Other Sports'
END
```

The calculation is valid.

1 Dependency ▾

Apply

OK

Male Dominat... 🔍 ⚙ ▾

☐ (All)
☒ Male-Dominated
☐ Other Sports

Season
☐ (All)
☒ Summer
☐ Winter

References

Dataset Link: Depinto, M. G. (2018, September 23). Let's discover more about the Olympic Games! Kaggle.

<https://www.kaggle.com/code/marcogdepinto/let-s-discover-more-about-the-olympic-games/notebook#7.-Medals-per-country>

Mural 23 Link:

<https://app.mural.co/t/datavisualizationfinalprojee5159/m/datavisualizationfinalprojee5159/1699563637854/1cc71ea8316fec9084fe918b1eabef52bdd8549e?sender=uc73c99f02de719af19555441>